Multiple Choice Questions on Data Types and Variables in Python

1. What is the data type of the following variable?
x = 10

A) float

B) int

C) str

D) bool

2. What will be the output of the following code?
x = 5.5
print(type(x))

A) int

B) float

C) str

D) bool

3. What will be the output of the following code?

x = "Hello"
y = 10
print(type(x) == type(y))

A) True

B) False

C) None

D) Error

4. What is the correct way to declare a string in Python?

A) x = 'Hello'

B) x = "Hello"

C) x = '''Hello'''

D) All of the above

5. What will be the output of the following code?

```
a = True
b = False
print(a + b)
```

A) 1

B) 0

C) True

D) False

6. What is the default data type of a variable assigned with a numeric value containing a decimal point?

A) int

B) float

C) double

D) complex

7. What will be the output of the following code?

```
x = 3 + 4j
print(type(x))
```

A) int

B) float

C) complex

D) str

8. What will be the output of the following code?

```
x = "10"
y = 2
print(x * y)
```

A) 20

B) 1010

C) 12

D) Error

9. Which of the following is a valid variable name in Python?

A) 2var

B) @var

C) var_name

D) _var

10. What will be the output of the following code?

```
x = None
```

print(type(x))

A) str

B) int

C) bool

D) NoneType

11. What is the output of bool("") in Python?

A) True
B) **False**
C) None
D) Error

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12. Which of the following is an immutable data type?

A) list
B) **tuple**
C) dictionary
D) set

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13. Which of the following statements correctly defines a variable in Python?

A) x : int = 10
B) x = 10
C) x, y = 5, "hello"
D) **All of the above**

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14. What is the correct syntax for defining a constant variable in Python?

A) const PI = 3.14
B) PI = 3.14
C) **PI = 3.14 # In uppercase to indicate a constant**
D) define PI 3.14

---

15. What will be the output of type([]) in Python?

A) **list**
B) tuple
C) dict
D) set

---

16. What will be the output of the following code?

```
x = 10

y = 10.5

print(type(x + y))
```

A) **float**
B) int
C) str
D) complex

---

17. What will be the output of the following code?

```
x = "5"

y = 2

print(x + str(y))
```

A) 7
B) **"52"**
C) Error
D) "25"

---

18. What will be the output of type({}) in Python?

A) list
B) **dict**
C) set
D) tuple

---

19. Which of the following is not a valid Python data type?

A) list
B) dictionary
C) array
D) **char**

---

20. What will be the output of the following code?

```
x = [1, 2, 3]

print(type(x))
```

A) tuple
B) dict
C) **list**
D) set

---

21. Which of the following statements is correct about Python variables?

A) Variables must be declared before use.
B) **Variables do not need explicit declaration.**
C) Variable type must be defined before assignment.
D) Python does not support variables.

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22. What will be the output of type(set())?

A) **set**
B) dict
C) list
D) tuple

---

23. What is the output of the following code?

```
x = (1,)

print(type(x))
```

A) list
B) **tuple**
C) set
D) int

---

24. What is the default data type of a variable assigned with a boolean value?

A) int
B) **bool**
C) str
D) float

---

25. What will be the output of the following code?

```
x = "Hello"

y = 'Hello'

print(x == y)
```

A) True
B) **True**
C) False
D) Error

---

26. What is the type of range(10) in Python?

A) list
B) tuple
C) **range**
D) generator

---

27. What will be the output of type({1, 2, 3})?

A) **set**
B) dict
C) list
D) tuple

---

28. What will be the output of type(True + 5)?

A) **int**
B) float
C) bool
D) str

---

29. Which of the following is a mutable data type?

A) tuple
B) **list**
C) string
D) int

---

30. What is the correct way to delete a variable in Python?

A) drop var_name
B) **del var_name**
C) delete var_name
D) remove var_name

---

31. What will be the output of the following code?

```
x = "Python"

print(len(x))
```

A) **6**
B) 5
C) Error
D) None

---

32. Which of the following is used to check the data type of a variable?

A) checktype()
B) typeof()
C) **type()**
D) datatype()

---

33. What will be the output of type(10/2) in Python?

A) int
B) **float**
C) double
D) complex

---

34. What will be the output of the following code?

```
x = {1, 2, 3, 3, 2}

print(len(x))
```

A) 5
B) **3**
C) 2
D) 4

---

35. Which of the following is true for Python lists?

A) Lists are immutable
B) **Lists are mutable**
C) Lists store only numeric values
D) Lists store only string values

---

36. What is the data type of the variable x in the following code?

```
x = (1, 2, "hello", 4.5)
```

A) list
B) **tuple**
C) set
D) dict

---

37. What will be the output of the following code?

```
x = 5

y = "Hello"
```

```
print(str(x) + y)
```

A) **"5Hello"**
B) 5”Hello”
C) Error
D) None

---

38. What is the result of the following code?

```
x = 3.0

y = int(x)

print(y)
```

A) 3.0
B) **3**
C) Error
D) None

---

39. Which of the following statements is true?

A) Tuple is mutable
B) **Dictionary keys must be unique**
C) List keys must be unique
D) Set allows duplicate values

---

40. What will be the output of the following code?

```
x = [1, 2, 3]

y = x

y.append(4)

print(x)
```

A) [1, 2, 3]
B) **[1, 2, 3, 4]**
C) Error
D) None

---

41. What will be the output of the following code?

```
x = (1, 2, 3)

x[0] = 4
```

A) **Error**
B) (4, 2, 3)
C) (1, 2, 3)
D) None

---

42. What will be the output of type({}) in Python?

A) set
B) **dict**
C) tuple
D) list

---

43. What will be the output of the following code?

```
x = 10

print(isinstance(x, int))
```

A) **True**
B) False
C) Error
D) None

---

44. What will be the output of the following code?

```
x = "Python"

print(x[0])
```

A) n
B) **P**
C) Error
D) None

---

45. Which of the following is a valid dictionary declaration?

A) x = ["name": "Alice", "age": 25]
B) x = [1, 2, 3]
C) x = {"name": "Alice", "age": 25}
D) x = (1, 2, 3)

---

46. What will be the output of the following code?

```
x = "Python"

print(x[-1])
```

A) **n**
B) Error
C) P
D) None

---

47. What will be the output of the following code

```
x = [1, 2, 3]

print(x * 2)
```

A) **[1, 2, 3, 1, 2, 3]**
B) [1, 2, 3, 6]
C) Error
D) None

---

48. What will be the output of bool("False") in Python?

A) **True**
B) False
C) None
D) Error

---

49. What will be the output of type([]) == type({})?

A) **False**
B) True
C) Error
D) None

---

50. What will be the output of the following code?

```
x = [1, 2, 3]

y = x.copy()

y.append(4)

print(x)
```

A) **[1, 2, 3]**
B) [1, 2, 3, 4]
C) Error
D) None